MEMORANDUM

TO: Dr. Tina Han

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CC: ME32301 Lab Section 08- Harbin, Zachary; Alshaikh, Leen

DATE: March-15-2025

RE: Torsion of Circular Shafts: Lab 7

EXECUTIVE SUMMARY

The main goal of this experiment is to determine how material stiffness given by shear modulus G and geometric stiffness affect the behavior of homogeneous, linear elastic circular shafts under pure torsion. Geometric stiffness of a material depends on the area polar moment of inertia J and shaft length L. The shear modulus G of a material, sometimes referred to as Modulus of rigidity, is the measure of rigidity of a body. It is expressed as the ratio between the shear stress and shear strain [1]. J is based on the geometry of the shaft and it is also a measure of the shaft's geometric stiffness to torsional deformation [2]. J for a circular shaft is given by equation 1 where D is the diameter of the shaft. The tested specimens during this lab were PTFE (polytetrafluoroethylene) Teflon shafts of various dimensions, as shown in table 1.

$$J = \frac{\pi D^4}{32} \tag{1}$$

In this experiment, a torque was applied to each specimen and the recorded angle of twist ϕ together with its calculated J was used to estimate the shear modulus G of each specimen. The relationship between ϕ and G is defined as follows,

$$\phi = \frac{TL}{JG} \tag{2}$$

The specimens' length L and diameters D were measured using a ruler and calipers 1. Furthermore, the torque and force were balanced before the specimen was installed. The specimen was then in-

Test	Specimen	Maximum Twist [deg]	Recording Increment [deg]
1	1/4 in x 3.5 in	10	2
2	1/4 in x 5.5 in	15	3
3	3/16 in x 5.5 in	15	3
4	3/16 in x 9.5 in	20	4

Table 1: Test Specimen Data

serted and secured between the grips of the Instron UTM Machine, ensuring that the marking on the specimen aligned with the V-faced jaw grips. To minimize error in output data, proper alignment and uniform grip on the shaft were ensured. Subsequently, displacement, torque, force and rotation readings were zeroed. The team made sure that the force and torque readings did not exceed 10 N and 1 Nm before starting the test. The test was then initiated and the specimen was carefully observed during the duration of the test. Upon completion of the test, the test data was collected and the sample was removed. The experimental data was then used to estimate the shear modulus of each specimen. This was achieved by utilizing least-squares method to estimate G, since the T-ϕ relationship is a linear function of G [2]. J was calculated using equation 1 and units of Nm for T, mm for L, and radians for ϕ were used to avoid an ill-conditioned optimization problem. Furthermore, additional MATLAB operations were performed to determine shear modulus G. As Table 2 illustrates, it is evident that experimental G values for the specimens had a high percentage difference, ranging from 21.90% to 54.70%, when compared to their respective published values G_{pub}. This can be explained as a result of the experiment being prone to various errors. Some of the potential sources of error include misalignment and non-uniform grip pressure, securing specimen in such a way as to avoid excess torque or force before start of experiment, measuring incorrect specimen length, and other computational errors or mistakes during calculations. Additionally, it is important to note that all experimental G values were within the 95% Confidence Interval.

D [mm]	J [mm^4]	L [mm]	G [GPa]	CI [GPa]	G_{pub}	% diff
6.35	159.6	88.9	0.1276	[0.1275, 0.1276]	0.23	44.52
6.35	159.6	139.7	0.1408	[0.1407, 0.1409]	0.23	38.78
4.7625	50.5	139.7	0.1042	[0.1037, 0.1048]	0.23	54.70
4.7625	50.5	241.3	0.1796	[0.1796, 0.1797]	0.23	21.91

Table 2: Comparison of experimental and published shear moduli

RESULTS

The corresponding torque T for the 1/4 inch x 3.5 inch specimen is shown in the table below at angles increasing from 0 degrees to 10 degrees in increments of 2 degrees, with ϕ being the angle of twist.

ϕ [deg]	ϕ [rad]	T [Nm]
0	0	0
2	0.0349	0.0068
4	0.0698	0.0150
6	0.1047	0.0231
8	0.1396	0.0308
10	0.1745	0.0384

Table 3: Measured torque of 1/4 in. x 3.5 in. specimen

The corresponding torque T for the 1/4 inch x 5.5 inch specimen is shown in the table below at angles increasing from 0 degrees to 15 degrees in increments of 3 degrees, with ϕ being the angle of twist.

ϕ [deg]	ϕ [rad]	T [Nm]
0	0	0
3	0.0523	0.0094
6	0.1047	0.0179
9	0.1571	0.0263
12	0.2094	0.0343
15	0.2618	0.0422

Table 4: Measured torque of 1/4 in. x 5.5 in. specimen

The corresponding torque T for the 3/16 inch x 5.5 inch specimen is shown in the table below at angles increasing from 0 degrees to 15 degrees in increments of 3 degrees, with ϕ being the angle of twist.

ϕ [deg]	ϕ [rad]	T [Nm]
0	0	0
3	0.0523	0.0023
6	0.1047	0.0050
9	0.1571	0.0066
12	0.2094	0.0080
15	0.2618	0.0092

Table 5: Measured torque of 3/16 in. x 5.5 in. specimen

The corresponding torque T for the 3/16 inch x 9.5 inch specimen is shown in the table below at angles increasing from 0 degrees to 20 degrees in increments of 4 degrees, with ϕ being the angle of twist.

ϕ [deg]	ϕ [rad]	T [Nm]
0	0	0
4	0.0698	0.0026
8	0.1396	0.0054
12	0.2094	0.0080
16	0.2793	0.0104
20	0.3491	0.0129

Table 6: Measured torque of 3/16 in. x 9.5 in. specimen

Figure 1 shows the plots of torque T versus angle of twist ϕ for the two 1/4 inch diameter shafts. The shaft with a length of 3.5 inches has a steeper slope than the shaft with a length of 5.5 inches, which makes sense because its length is shorter.

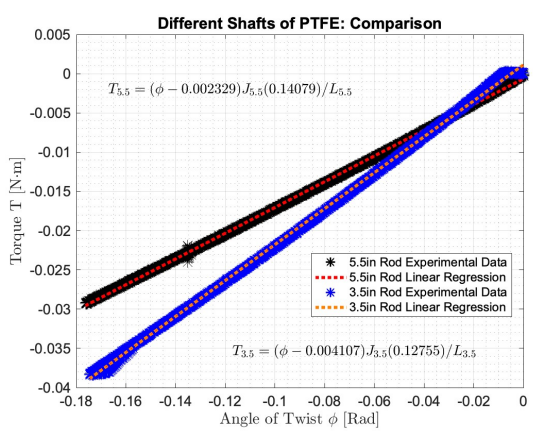

Figure 1: Comparison of two 1/4 in. diameter shafts

Figure 2 shows the plots of torque T versus angle of twist ϕ for the two 3/16 inch diameter shafts. The data for the 5.5 inch length shaft is slightly erroneous, making it slightly nonlinear. This is likely caused by the specimen slipping between the rigid holds during testing. The similar slopes indicate that the 9.5 inch length shaft has a higher shear modulus because it is divided by a larger number due to its longer length.

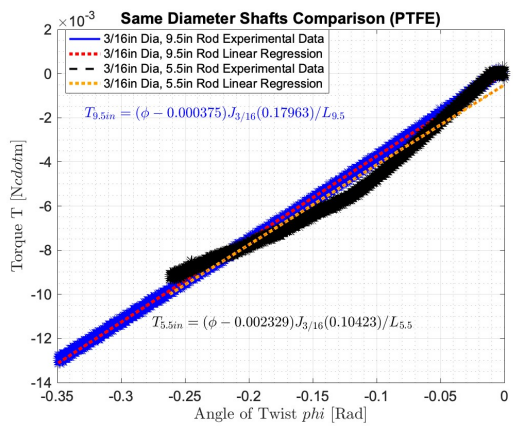

Figure 2: Comparison of two 3/16 in. diameter shafts

Figure 3 shows the plots of torque T versus angle of twist ϕ for the two 5.5 inch length shafts. The shaft with a diameter of 1/4 inch has a steeper slope than the shaft with a diameter of 3/16 inch, which matches with the fact that its polar moment of inertia is higher.

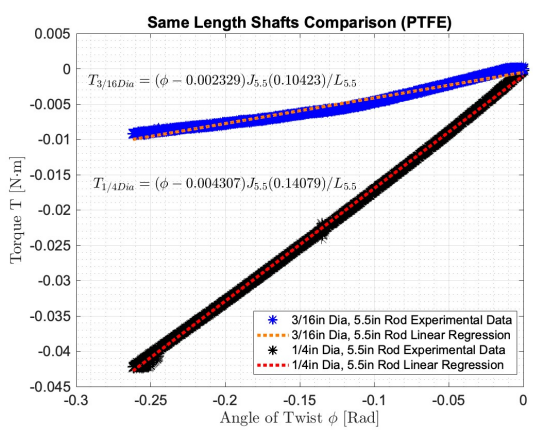

Figure 3: Comparison of two 5.5 in. length shafts

All the three plots were generated after the unloading path was removed from the collected data. The table below shows the difference between experimental and published shear moduli for the

different specimens, with D being diameter, J being the area polar moment of inertia, L being the length, G being the experimental shear modulus, CI being the 95% confidence interval, G_{pub} being the published shear modulus, and % diff being the percentage difference between the experimental and published shear moduli [3]. The G values were calculated using the $fmincon$ optimization function, starting from initial guesses. The given function shaft_model() was run after making appropriate changes to pre-written equations and referencing the collected data using readmatrix commands. The process was repeated with bootstrapped samples to compute 95% Confidence Intervals for G. Moreover, the J values were calculated using Equation 1.

D [mm]	J [mm^4]	L [mm]	G [GPa]	CI [GPa]	G_{pub}	% diff
6.35	159.6	88.9	0.1276	[0.1275, 0.1276]	0.23	44.52
6.35	159.6	139.7	0.1408	[0.1407, 0.1409]	0.23	38.78
4.7625	50.5	139.7	0.1042	[0.1037, 0.1048]	0.23	54.70
4.7625	50.5	241.3	0.1796	[0.1796, 0.1797]	0.23	21.91

Table 7: Comparison of experimental and published shear moduli

As shown in Table 7, the difference between the experimental and published shear moduli goes down as length increases and diameter decreases, which the exception of the specimen whose test experienced some error.

CONCLUSIONS

The goal of this lab is to validate the theory of torsion—that is, to determine the influence of shear modulus G, polar moment of inertia J, and shaft length L on the behavior of homogeneous, linear elastic circular shafts under pure torsion.

Teflon shafts of various dimensions were tested on an Instron UTM machine with a Torsion Add-On, which applied and measured an angle of twist, while the resulting torque was recorded at multiple angle increments by an inbuilt sensor. The length of each specimen was measured, and the polar area moment of inertia was calculated using Equation 1. Using the information from these variables, a linear regression model was employed to estimate the shear modulus using the least-squares method. This shear modulus was then compared to published values, and the percent

difference as well as confidence intervals were calculated, as shown in Table 7.

The experimental G values for every beam differed significantly from the published Teflon G value of 0.23. As seen in Table 7, the largest percent difference was 54.70% for the 3/16 in x 5.5 in specimen, which had a calculated G of 0.1042. The second-largest percent error was 44.52% for the 1/4 in x 3.5 in specimen, which had a calculated G of 0.1276. The third-largest percent error was 38.78% for the 1/4 in x 5.5 in specimen, which had a calculated G of 0.1408. The smallest percent error was 21.91% for the 3/16 in x 9.5 in specimen, which had a calculated G of 0.1796. All these calculated G values were within their 95% confidence intervals.

These large differences between the calculated and theoretical values could have resulted from various sources, including: misalignment when inserting the specimens into the grips of the Instron UTM machine, non-uniform grip pressure from the Instron UTM machine, worn-down specimens whose diameters do not align with the posted values (which would cause errors in the area polar moment of inertia calculation), measuring incorrect specimen length, incorrectly recording the angle of twist readings (which would then cause errors in the linear regression calculations), as well as other computational errors or mistakes during calculations.

Another factor that would significantly impact the calculated G values is the accuracy of the Instron Machine and Torsion Add-On, which, as stated earlier, applied and measured an angle of twist and used an inbuilt sensor to measure the respective torque measurements. Inaccurate measurements for either the angle of twist or torque would affect the linear regression model since both values are used in the least-squares method to calculate the shear modulus G.

Additionally, it is important to correctly measure the length and diameter of the circular shaft specimens, as these two values directly impact the torque under the same angle of twist. Equation 2, rearranged for torque, can be seen in Equation 3 below. T is inversely related to L, so a longer specimen would create a higher torque and vice versa. As seen in Equation 1, J is inversely proportional to the diameter of the specimen to the fourth power, meaning that increasing the diameter also increases J. Since torque is directly proportional to J, as shown in Equation 3, torque would

also increase. The opposite is also true: if the diameter decreases, J decreases, and T decreases.

$$T = \frac{\phi J G}{L} \tag{3}$$

Measuring the dimensions (length and diameter) of the specimens incorrectly would not affect the measurements of torque under the same angle of twist since the UTM machine does not take those values into account when sensing the angle of twist and torque. However, this would affect the linear regression model later on since both values are used in the least-squares method, which is used to calculate the shear modulus G.

It is also important to ensure that the assumptions that underlie the Torsional formulas used in this lab were reasonable. The first assumption that was made is that the specimens were straight, slender members with a circular cross-section. This was validated at the beginning of the experiment upon inspection of the member, making it a reasonable assumption for this experiment. Another assumption made in this lab is that the material is isotropic, meaning it has constant material properties. This includes a constant shear modulus G, as well as a constant diameter of the beam, yielding a constant area polar moment of inertia. The shear modulus assumption holds with the Teflon specimens used since they were made of the same material throughout. Additionally, although the samples were not measured in multiple sections throughout their length and there could have been slight variations in the diameter, those would be minimal, making the isotropic assumption reasonable for this lab.

Another assumption that was made was that the axis of the specimen remained straight, with the cross-sections of the specimen perpendicular to the axis before and after deformation. This was enforced during this lab when inserting the specimen into the UTM machine, aligning the markings on the specimen, and aligning the specimen with the V-faced jaw grips. The last assumption that was made is that there were only small deformations in the samples, meaning a small angle of twist. The smallest angle of twist was for the first specimen at 10 degrees, and the highest was 20 degrees.

This does not qualify as a small angle of twist, making this assumption invalid and thus potentially introducing error into the lab. This could have been a big contributor as to why the shear modulus percent errors were so high.

REFERENCE LIST

[1] BYJU'S. "Shear modulus (modulus of rigidity) - definition, formula, units, examples." (2022), [Online]. Available: `https://byjus.com/physics/shear-modulus-elastic-moduli/` (visited on 03/17/2025).

[2] M. Gonzalez. "Me 32301 - mechanics of materials laboratory." (2023).

[3] designerdata. "Polytetrafluoroethylene (teflon)." (), [Online]. Available: `https://designerdata.nl/materials/plastics/thermo-plastics/polytetrafluoroethylene-(teflon)` (visited on 03/15/2025).

[4] OpenAI. "ChatGPT (March 19th, 2025 Version)," [Online]. Available: https://openai.com/chat-gpt (visited on 03/19/2025).

DATA SHEETS

The following 4 screenshots are from the MS Excel file that contained the rotation and torque values for the different shafts used in lab. The data was generated by performing a torsion test on PTFE shafts of varying lengths and diameters. This data was used to estimate the Shear modulus values and the plots.

	A	B	C	D	E
1	Time	Displacement	Force	Rotation	Torque
2	(s)	(mm)	(kN)	(deg)	(N-m)
3	0	1.2303	-0.0007	0	0
4	0.02	1.2303	-0.0007	0	0.0001
5	0.04	1.2303	-0.0006	0	0.0001
6	0.06	1.2303	-0.0006	0	0.0001
7	0.08	1.2303	-0.0006	0	0
8	0.1	1.2303	-0.0006	0	0
9	0.12	1.2303	-0.0006	0	0
10	0.14	1.2303	-0.0007	0	0
11	0.16	1.2303	-0.0007	0	0
12	0.18	1.2303	-0.0007	0	0.0001
13	0.2	1.2303	-0.0007	0	0.0001
14	0.22	1.2303	-0.0006	0	0.0001
15	0.24	1.2303	-0.0006	0	0
16	0.26	1.2303	-0.0006	-0.0002	0.0001
17	0.28	1.2303	-0.0006	-0.0006	0.0001
18	0.3	1.2303	-0.0006	-0.0009	0.0001
19	0.32	1.2303	-0.0007	-0.0009	0.0001
20	0.34	1.2303	-0.0007	-0.0009	0.0001
21	0.36	1.2303	-0.0007	-0.0009	0.0001
22	0.38	1.2303	-0.0006	-0.0009	0.0001
23	0.4	1.2303	-0.0006	-0.001	0
24	0.42	1.2303	-0.0006	-0.0015	0
25	0.44	1.2303	-0.0006	-0.0022	0.0001
26	0.46	1.2303	-0.0006	-0.0029	0
27	0.48	1.2303	-0.0006	-0.0038	0

Figure 4: Datasheet for 1/4 in Diameter, 3.5 in PTFE Shaft

	A	B	C	D	E
1	Time	Displacement	Force	Rotation	Torque
2	(s)	(mm)	(kN)	(deg)	(N-m)
3	0	0	0	0	0
4	0.02	0	0	0	-0.0001
5	0.04	0	0	0	-0.0001
6	0.06	0	0	0	-0.0001
7	0.08	0	0	0	-0.0001
8	0.1	0	0	0	0
9	0.12	0	0	0	0
10	0.14	0	0	0	-0.0001
11	0.16	0	0	0	-0.0001
12	0.18	0	0	0	-0.0003
13	0.2	0	0	0	-0.0002
14	0.22	0	0	0	-0.0001
15	0.24	0	0	0	-0.0001
16	0.26	0	0	0	-0.0001
17	0.28	0	0	0	-0.0002
18	0.3	0	0	0	-0.0002
19	0.32	0	0	0	-0.0002
20	0.34	0	0	0	-0.0001
21	0.36	0	0	0	-0.0001
22	0.38	0	0	0	-0.0001
23	0.4	0	0	0	-0.0001
24	0.42	0	0	0	-0.0002
25	0.44	0	0	0	-0.0001
26	0.46	0	0	0	0
27	0.48	0	0	-0.0002	0

Figure 5: Datasheet for 1/4 in Diameter, 5.5 in PTFE Shaft

	A	B	C	D	E
1	Time	Displacement	Force	Rotation	Torque
2	(s)	(mm)	(kN)	(deg)	(N-m)
3	0	-1.4973	-0.0013	0	0.0001
4	0.02	-1.4973	-0.0013	0	0
5	0.04	-1.4973	-0.0014	-0.0001	0
6	0.06	-1.4973	-0.0014	-0.0006	0
7	0.08	-1.4973	-0.0013	-0.0009	0.0001
8	0.1	-1.4973	-0.0014	-0.0009	0
9	0.12	-1.4973	-0.0013	-0.0009	0
10	0.14	-1.4973	-0.0013	-0.0009	-0.0001
11	0.16	-1.4973	-0.0014	-0.0009	0
12	0.18	-1.4973	-0.0014	-0.001	0
13	0.2	-1.4973	-0.0013	-0.0015	0
14	0.22	-1.4973	-0.0013	-0.0018	0.0001
15	0.24	-1.4973	-0.0013	-0.002	0.0001
16	0.26	-1.4973	-0.0014	-0.0025	0
17	0.28	-1.4973	-0.0014	-0.003	0
18	0.3	-1.4973	-0.0013	-0.0035	0.0001
19	0.32	-1.4973	-0.0014	-0.0042	0.0001
20	0.34	-1.4973	-0.0013	-0.0049	0
21	0.36	-1.4973	-0.0014	-0.0057	-0.0001
22	0.38	-1.4973	-0.0014	-0.0065	-0.0001
23	0.4	-1.4973	-0.0014	-0.0073	0
24	0.42	-1.4973	-0.0013	-0.0082	0
25	0.44	-1.4973	-0.0013	-0.009	0.0001
26	0.46	-1.4973	-0.0013	-0.01	0
27	0.48	-1.4973	-0.0013	-0.0109	0

Figure 6: Datasheet for 3/16 in Diameter, 5.5 in PTFE Shaft

	A	B	C	D	E
1	Time	Displacemen	Force	Rotation	Torque
2	(s)	(mm)	(kN)	(deg)	(N-m)
3	0	0	-0.0002	0	0
4	0.02	0	-0.0002	0	0.0001
5	0.04	0	-0.0002	0	0
6	0.06	0	-0.0002	0	0
7	0.08	0	-0.0002	0	0
8	0.1	0	-0.0002	0	0.0001
9	0.12	0	-0.0002	0	0.0001
10	0.14	0	-0.0002	0	0
11	0.16	0	-0.0002	0	0
12	0.18	0	-0.0002	0	0
13	0.2	0	-0.0002	0	-0.0001
14	0.22	0	-0.0002	0	-0.0001
15	0.24	0	-0.0002	0	0.0001
16	0.26	0	-0.0002	0	0.0001
17	0.28	0	-0.0001	0	0.0001
18	0.3	0	-0.0001	0	0
19	0.32	0	-0.0001	0	0.0001
20	0.34	0	-0.0002	0	0.0001
21	0.36	0	-0.0002	0	0.0001
22	0.38	0	-0.0002	0	0
23	0.4	0	-0.0002	0	0
24	0.42	0	-0.0002	-0.0004	-0.0001
25	0.44	0	-0.0002	-0.0008	0.0001
26	0.46	0	-0.0001	-0.001	0
27	0.48	0	-0.0001	-0.0016	0

Figure 7: Datasheet for 3/16 in Diameter, 9.5 in PTFE Shaft

APPENDIX

The following code was used to calculate the Shear Modulus (G) and displacements for the different shapes and sizes of the PTFE shafts used in the lab. The code was further used to calculate the 95% Confidence Intervals for these two quantities.

```
function shaft_model()
    % In the space below, import the data sheet(s) obtained from the Instron UTM.
    % Hint: This can be done through a function called "readmatrix", which
    % has the following documentation: https://www.mathworks.com/help/matlab/ref/readmatrix.html

    torque_alum_three_sixteen = readmatrix("3_16_D_9_5_L_Speciman_Test1.csv");

    % Referencing the imported data sheet, define the list of values for
    % torque and angle of twist.
    T = torque_alum_three_sixteen(1:end, 5); % Torque [N*m]
    phi = torque_alum_three_sixteen(1:end, 4)*(pi/180); % Angle of twist [radians]
    % Calculate the area polar moment of intertia and define the length of
    % the specimen.
    J = pi*(4.7625)^4/(32); % Area polar moment of intertia [mm^4]
    L = 241.3; % Length of the shaft [mm]
    % Defining the number of data points and number of bootstrap samples
    n = length(T); % number of data points
    numBootstraps = 1000; % number of bootstrap samples
    % Initial guess for shear modulus G and error in angle of twist phiE
    G0 = 1;
    phiE0 = 0;
    % Define objective function to find optimum values
    objfun = @(x) sum((T - (J*x(1)/L) .* (phi-x(2))).^2);
    % Define objective function for bootstrapping
```

```
objfun_bootstrap = @(x,T,J,L,phi) sum((T - (J*x(1)/L) .* (phi-x(2))).^2);
% Define constraints for G (G > 0)
options = optimoptions(@fmincon,'Algorithm','interior-point');
lb = [0, -Inf]; % lower bound
ub = [Inf, Inf]; % upper bound
% Call fmincon to minimize objective function
x = fmincon(objfun, [G0, phiE0], [], [], [], [], lb, ub, [], options);
% Initialize array to hold bootstrap estimates
bootstrap_estimates = zeros(numBootstraps,2);
% Perform bootstrap estimation
for i = 1:numBootstraps
    % Generate bootstrap sample
    ind = randi(n,n,1);
    T_b = T(ind);
    J_b = J;
    L_b = L;
    phi_b = phi(ind);
    % Call fmincon to minimize objective function for bootstrap sample
    x_b = fmincon(@(x) objfun_bootstrap(x,T_b,J_b,L_b,phi_b),...
    [G0,phiE0],[],[],[],[],lb,ub,[],options);
    bootstrap_estimates(i,:) = x_b;
end
% Compute confidence intervals
alpha = 0.05; % 95% confidence interval
G_conf_int = quantile(bootstrap_estimates(:,1),[alpha/2, 1-alpha/2]);
phiE_conf_int = quantile(bootstrap_estimates(:,2),[alpha/2, 1-alpha/2]);
% Display the optimal value of G and phiE
disp(['Optimal value of G: ', num2str(x(1)), ' GPa']);
disp(['Optimal value of phiE: ', num2str(x(2)), ' radians']);
% Display the confidence intervals
disp(['95% confidence interval for G: [', num2str(G_conf_int(1)), ', ',
num2str(G_conf_int(2)), ']', ' GPa'])
```

```
    disp(['95% confidence interval for phiE: [', num2str(phiE_conf_int(1)), ',
    ', num2str(phiE_conf_int(2)), ']', ' radians'])
end
```

The following code was used to make different plots such as comparison of different diameters, shapes, and sizes. The same code was rerun 3 times by mainly changing the readmatrix command to input the correct files. Furthermore, with minor changes in the code, such as changing legends, equation values, and formatting, the three plots shown in the Results section were produced.

```
% data extraction from excel
lab7_1 = readmatrix("2025_SP_Section08_Team4_Lab07_0.25x5.5_13.xlsx");
lab7_2 = readmatrix("Lab7Machine1Group4Section8_1.xlsx");
% variable setting for both rotations
rotation1 = lab7_1(1:10534, 4) * (pi / 180);
rotation2 = lab7_2(1:10534, 4) * (pi / 180);

% variable setting for both torques
torque1 = lab7_1(1:10534, 5);
torque2 = lab7_2(1:10534, 5);

% code to generate plot for experimental data
figure(1)
hold on
grid on
grid minor
box on

plot(rotation1, torque1, '-', 'LineWidth', 2);
plot(rotation2, torque2, '--', 'LineWidth', 2);

% linear regression plot code
mdl_lab7_1 = fitlm(rotation1, torque1, 'VarNames', {'rotation1', 'torque1'});
mdl_lab7_2 = fitlm(rotation2, torque2, 'VarNames', {'rotation2', 'torque2'});
```

```

% commands for the lines in the plot
mdl_plot_lab7_1 = plot(mdl_lab7_1);
mdl_plot_lab7_1(1).MarkerSize = 10;
mdl_plot_lab7_1(1).Marker = "*";
mdl_plot_lab7_1(1).Color = "k";
mdl_plot_lab7_1(1).MarkerFaceColor = "auto";
% used from lab 1
mdl_plot_lab7_1(2).LineStyle = ":";
mdl_plot_lab7_1(2).LineWidth = 3;

% commands for the lines in the plot
mdl_plot_lab7_2 = plot(mdl_lab7_2);
mdl_plot_lab7_2(1).MarkerSize = 10;
mdl_plot_lab7_2(1).Marker = "*";
mdl_plot_lab7_2(1).Color = "b";
mdl_plot_lab7_2(1).MarkerFaceColor = "auto";
% used from lab 1
mdl_plot_lab7_2(2).LineStyle = ":";
mdl_plot_lab7_2(2).LineWidth = 3;


% Define and annotate regression equation for 3.5in
Eq_text_lab7_1 = sprintf('$T_{3.5} = (\\phi - %.6f)J_{3.5}(%.5f)/L_{3.5}$', ...
        0.004107, 0.12755);
Eq_Anno_lab7_1 = text(mean(rotation1) + 0.02, min(torque1) - 0.005, Eq_text_lab7_1, ...
                    'Interpreter', 'latex', 'FontSize', 14, 'Color', 'k', 'HorizontalAlignment', 'center');

% Define and annotate regression equation for 5.5in
Eq_text_lab7_2 = sprintf('$T_{5.5} = (\\phi - %.6f)J_{5.5}(%.5f)/L_{5.5}$',
```

```
    ...
        0.002329, 0.14079);
Eq_Anno_lab7_2 = text(mean(rotation2) - 0.03, max(torque2) - 0.002,
    Eq_text_lab7_2, ...
                        'Interpreter', 'latex', 'FontSize', 14, 'Color', 'r', '
    HorizontalAlignment', 'center');



xlabel('Angle of Twist $\phi$ [Rad]', 'Interpreter', 'latex', 'FontSize', 14);
ylabel('Torque T [N$\cdot$m]', 'Interpreter', 'latex', 'FontSize', 14);
title("Different Shafts of PTFE: Comparison ", 'FontSize', 16);

% more formatting such as legends
legend({'3.5in Rod Experimental Data', '3.5in Rod Linear Regression', '5.5in
    Rod Experimental Data', '5.5in Rod Linear Regression'}, ...
        'FontSize', 12, 'Location', 'best');

% setting font features such as name and size
ax = gca;
ax.FontName = 'Arial';
ax.FontSize = 14;

hold off
```